PRO MINDS

Algebra 1 — Arithmetic & Geometric Sequences

Finding terms from a rule, recursive and explicit formulas, sequence types, error analysis, and applications · 16 questions

Name: _____ Date: _____

Free practice worksheet — show your work in the space under each question, then check yourself with the answer key.

Question 1 · Terms of a sequence

A(n) = 4 + (n − 1)(−5)

Find the third, fifth, and tenth terms of the sequence.

Question 2 · Terms of a sequence

A(n) = 1.4 + (n − 1)(3)

Find the third, fifth, and tenth terms of the sequence.

Question 3 · Arithmetic sequences · Recursive & explicit formulas

5, 13, 21, 29, ...

Recursive formula:

Explicit formula:

Question 4 · Arithmetic sequences · Recursive & explicit formulas

51, 47, 43, 39, ...

Recursive formula:

Explicit formula:

Question 5 · Arithmetic sequences · Recursive & explicit formulas

0.2, 0.5, 0.8, 1.1, ...

Recursive formula:

Explicit formula:

Question 6 · Arithmetic sequences · Recursive & explicit formulas

1.6, 0.8, 0, −0.8, ...

Recursive formula:

Explicit formula:

Question 7 · Error analysis

Your friend writes A(8) = 3 + (8)(5) as an explicit formula for finding the eighth term of the arithmetic sequence 3, 8, 13, 18, ...

Describe and correct your friend's error.

Question 8 · Identify the sequence

5, 6, 8, 11, ...

Identify the sequence as arithmetic, geometric, or neither.

Question 9 · Identify the sequence

1, 1/2, 1/4, 1/8, ...

Identify the sequence as arithmetic, geometric, or neither.

Question 10 · Identify the sequence

1, 4, 7, 10, ...

Identify the sequence as arithmetic, geometric, or neither.

Question 11 · Geometric sequences · Recursive & explicit formulas

5, 15, 45, 135, ...

Recursive formula:

Explicit formula:

Question 12 · Geometric sequences · Recursive & explicit formulas

−2, −12, −72, −432, ...

Recursive formula:

Explicit formula:

Question 13 · Geometric sequences · Recursive & explicit formulas

500, 100, 20, 4, ...

Recursive formula:

Explicit formula:

Question 14 · Geometric sequences · Recursive & explicit formulas

75, 15, 3, 3/5, ...

Recursive formula:

Explicit formula:

Question 15 · Application

Suppose you want to make a reduced copy of a photograph that has an actual length of 8 inches. Each time you press the reduce button on the copier, the copy is reduced by 12%. What is the length of the photograph's copy if you press the reduce button 5 times? Round to the nearest hundredth.

Question 16 · Application

A geometric sequence has an initial value of 3 and a common ratio of 2.

Write a function to represent this sequence.